Stochastic Interpolants [1]

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1 Introduction

Contemporary approaches to generate data from a target distribution do so by continuously transforming samples from a simple base distribution (e.g., Gaussian noise) into the target distribution using ordinary or stochastic differential equations (ODEs/SDEs). A major breakthrough in this area was score-based diffusion **(Song et al., 2021b)**, which maps data to noise and learns to reverse this process by estimating the score function and running an SDE. Although effective, these methods typically rely on infinite-time dynamics (truncated in practice) and often assume a Gaussian endpoint. It was then shown that ODE-based methods (flow-matching/rectified flow/normalizing flows) could perform just as well, if not better than SBDMs **(Lipman et al., 2022)**. Moreover, these ODE-based techniques had the additional advantage of relying on cleaner, deterministc flows.

The proposal of *stochastic interpolants* **(Albergo and Vanden-Eijnden, 2023)** unified the two approaches and fulfilled the need for alternative methods that (i) operate between arbitrary distributions, and (ii) achieve exact transport over a finite time interval. In particular, it is a continuous-time process that connects samples from an initial distribution ρ_0 to a target distribution ρ_1 by inducing a family of intermediate distributions and can be realized through multiple dynamics, including both ODEs and forward/backward SDEs.

A key insight is that the drift terms of these dynamics can be learned from data by solving a least-squares regression problem. Once learned, these dynamics enable exact transport of samples between distributions over finite time, providing a flexible and theoretically grounded alternative to previous approaches.

2 Formulation

Stochastic Interpolant. Let ρ_0 and ρ_1 be two probability distributions on $\mathbb{R}^d$. A stochastic interpolant is defined as the stochastic process

$$x_t = I(t, x_0, x_1) + \gamma(t)z, \quad t \in [0, 1], \tag{1}$$

where

- $I(t, x_0, x_1) \in C^2([0, 1], C^2(\mathbb{R}^d \times \mathbb{R}^d)^d)$ satisfying the boundary conditions

  $$I(0, x_0, x_1) = x_0, \quad \text{and} \quad I(1, x_0, x_1) = x_1.$$

  Additionally, there exists a constant $C > 0$ such that for all $t \in [0, 1]$ and $x_0, x_1 \in \mathbb{R}^d$,

  $$\|\partial_t I(t, x_0, x_1)\| \leq C\|x_0 - x_1\|. \tag{2}$$

- The function $\gamma(t)$ is a time-dependent scaling factor such that $\gamma^2(t) \in C^2([0, 1])$, with boundary conditions

  $$\gamma(0) = \gamma(1) = 0, \quad \text{and} \quad \gamma(t) > 0 \text{ for } t \in (0, 1).$$

  In particular, $|\frac{d}{dt}\big(\gamma^2(t)\big)|$ is bounded.

- The pair (x_0, x_1) is sampled from a joint distribution ν whose marginals are ρ_0 and ρ_1.

- Finally, $z \sim \mathcal{N}(0, I_d)$ independent of (x_0, x_1).

The latent variable $\gamma(t)z$ will serve to provide us with nice smoothness properties.
Before we move on, we assume additonally that the densities ρ_0 and ρ_1 are strictly positive functions in $C^2(\mathbb{R}^d)$ and satisfy

$$\int_{\mathbb{R}^d} |\nabla \log \rho_0(x)|^2 \rho_0(x)\, dx < \infty, \quad \int_{\mathbb{R}^d} |\nabla \log \rho_1(x)|^2 \rho_1(x)\, dx < \infty. \tag{3}$$

Moreover, assume there exists $M_1, M_2 < \infty$ such that $\mathbb{E}\big[|\partial_t I(t, x_0, x_1)|^4\big] \leq M_1$ and $\mathbb{E}\Big[|\partial_t^2 I(t, x_0, x_1)|^2\Big] \leq M_2$ for all $t \in [0, 1]$, where the expectation is over $(x_0, x_1) \sim \nu$.

3 Transport Equation and ODE point of view

It can be shown that the probability distribution of the stochastic interpolant x_t defined in (1) is absolutely continuous with respect to the Lebesgue measure for all $t \in [0,1]$, and the time-dependent density $\rho(t,x)$ satisfies $\rho(0,x) = \rho_0(x)$, $\rho(1,x) = \rho_1(x)$, and $\rho \in C^1\big([0,1]; C^p(\mathbb{R}^d)\big)$ for any $p \in \mathbb{N}$, with $\rho(t,x) > 0$ for all $(t,x) \in [0,1] \times \mathbb{R}^d$. Moreover, $\rho(t,x)$ satisfies the transport equation

$$\partial_t \rho + \nabla \cdot (b\rho) = 0, \tag{4}$$

where the velocity field $b(t,x)$ is defined by

$$b(t,x) = \mathbb{E}[\dot{x}_t \mid x_t = x] = \mathbb{E}\big[\partial_t I(t, x_0, x_1) + \dot{\gamma}(t) z \,\big|\, x_t = x\big]. \tag{5}$$

This velocity satisfies $b \in C^0\big([0,1]; (C^p(\mathbb{R}^d))^d\big)$ for any $p \in \mathbb{N}$, and, for all $t \in [0,1]$,

$$\int_{\mathbb{R}^d} |b(t,x)|^2 \rho(t,x)\, dx < \infty. \tag{6}$$

All of these regularity properties follow from $\gamma(t) > 0$, as the latent Gaussian imposes exponentially decaying fourier modes for the density $\rho(t,x)$ and subsequently the velocity $b(t,x)$ via analysis of the characteristic function of x_t.

Thus the law of x_t is equivalent to the law of the stochastic process X_t that solves the following probability flow ODE

$$\frac{d}{dt} X_t = b(t, X_t), \tag{7}$$

which can be solved either forward in time with initial condition $X_{t=0} \sim \rho_0$, or backward in time with terminal condition $X_{t=1} \sim \rho_1$. This allows us to generate samples from the terminal distribution by pushing any initial samples through (7).

Using this ODE-based method requires learning the velocity field $b(t,x)$ by designing an ERM using (5). That said, the minimization is usually imperfect, and so it is important to check the stability of the transport equation under perturbed velocity. As it turns out, the result is unfavorable. In fact, for two different velocities $b, \hat{b}$ and their corresponding solutions $\rho, \hat{\rho}$ satisfy:

$$\mathrm{KL}\big(\rho(1) \,\|\, \hat{\rho}(1)\big) = \int_0^1 \int_{\mathbb{R}^d} (\nabla \log \hat{\rho}(t,x) - \nabla \log \rho(t,x)) \cdot \big(\hat{b}(t,x) - b(t,x)\big)\, \rho(t,x)\, dx\, dt. \tag{8}$$

Proof. Consider

$$\begin{aligned}
\frac{d}{dt}\mathrm{KL}\big(\rho(t)\,\|\,\hat\rho(t)\big) = \frac{d}{dt}\int_{\mathbb{R}^d}\rho\log\frac{\rho}{\hat\rho}\,dx &= \int_{\mathbb{R}^d}\left(\partial_t\rho\log\frac{\rho}{\hat\rho} + \rho\,\partial_t\log\frac{\rho}{\hat\rho}\right)dx \\
&= \int_{\mathbb{R}^d}\left(\partial_t\rho\log\frac{\rho}{\hat\rho} + \partial_t\rho - \frac{\rho}{\hat\rho}\partial_t\hat\rho\right)dx
\end{aligned}$$

Using the transport equations we obtain

$$\begin{aligned}
\frac{d}{dt}\mathrm{KL}\big(\rho\,\|\,\hat\rho\big) &= \int_{\mathbb{R}^d}\left[-\log\frac{\rho}{\hat\rho}\nabla\cdot(b\rho) - \nabla\cdot(b\rho) + \frac{\rho}{\hat\rho}\nabla\cdot(\hat b\hat\rho)\right]dx \\
&\overset{i.b.p}{=} \int_{\mathbb{R}^d}\left[\nabla\log\frac{\rho}{\hat\rho}\cdot b\rho - \nabla\left(\frac{\rho}{\hat\rho}\right)\cdot\hat b\hat\rho\right]dx \\
&= \int_{\mathbb{R}^d}(\nabla\log\rho - \nabla\log\hat\rho)\cdot b\rho\,dx + \int_{\mathbb{R}^d}(\nabla\log\hat\rho - \nabla\log\rho)\cdot\hat b\rho\,dx \\
&= \int_{\mathbb{R}^d}(\nabla\log\hat\rho - \nabla\log\rho)\cdot(\hat b - b)\,\rho\,dx.
\end{aligned}$$

Integrating in time from 0 to 1 gives us the result. □

This implies that matching the velocity alone is generally insufficient to guarantee convergence. Nevertheless, it can be shown that the SDE formulation does not have this problem.

4 Fokker Planck Equations and SDE point of view

Note that the regularity and positivity of ρ imples that the score $s = \nabla\log\rho \in C^1\big([0,1];C^p(\mathbb{R}^d)\big)$ for any $p\in\mathbb{N}$. A similar analysis as before of characteristic functions shows that

$$s(t,x) = \gamma^{-1}(t)\mathbb{E}(z|x_t = x), \tag{9}$$

and that for all $t\in[0,1]$,

$$\int_{\mathbb{R}^d}|s(t,x)|^2\rho(t,x)\,dx < \infty. \tag{10}$$

This access to the score allows one to generalize the transport equation from (4) to a family of (forward and backward) Fokker-Planck equations indexed by a positive time-dependent scalar $\epsilon(t)\in C^0([0,1])$ and given by

$$\partial_t\rho + \nabla\cdot(b_F\rho) = \epsilon(t)\,\Delta\rho, \quad \rho(0) = \rho_0, \tag{11}$$

where the forward drift is defined as

$$b_F(t,x) = b(t,x) + \epsilon(t)\,s(t,x), \tag{12}$$

and

$$\partial_t \rho + \nabla \cdot (b_B \rho) = -\epsilon(t)\, \Delta\rho, \quad \rho(1) = \rho_1, \tag{13}$$

where the backward drift is defined as

$$b_B(t,x) = b(t,x) - \epsilon(t)\, s(t,x). \tag{14}$$

Both are well-posed when solved forward/backward in time and each satisfies $\rho(0) = \rho_0$ and $\rho(1) = \rho_1$. Moreover, this imples that the law of x_t coincides with the laws of the stochastic processes X_t^F and X_t^B that solve the corresponding (forward and backward) SDEs

$$dX_t^F = b_F(t, X_t^F)\, dt + \sqrt{2\epsilon(t)}\, dW_t, \tag{15}$$

solved forward in time from the initial condition $X_{t=0}^F \sim \rho_0$, independent of W, and

$$dX_t^B = b_B(t, X_t^B)\, dt + \sqrt{2\epsilon(t)}\, dW_t^B, \quad W_t^B = -W_{1-t}, \tag{16}$$

solved backward in time from the final condition $X_{t=1}^B \sim \rho_1$, independent of W^B. Similar to the ODE case, this allows us to push samples from any initial distribution to the other via either (15) or (16). Note that setting $\epsilon = 0$ recovers the ODE case.

Using this SDE based method requires learning not only the drift $b(t,x)$, but also the score $s(t,x)$ via an ERM using (9). This time though, the control of the propagation of the statistical error is more favorable, i.e, for different $b, \hat{b}$ and $s, \hat{s}$ and corresponding solutions to the FPEs $\rho, \hat{\rho}$ satisfy (only forward shown below but similar result holds for backward FPE):

$$\mathrm{KL}\big(\rho(1) \,\|\, \hat{\rho}(1)\big) \le \frac{1}{4\epsilon} \int_0^1 \int_{\mathbb{R}^d} \Big|\hat{b}_F(t,x) - b_F(t,x)\Big|^2 \rho(t,x)\, dx\, dt, \tag{17}$$

where the RHS is minimized when b and s are learnt (recall $b_F = b + \epsilon s$ and so $\hat{b}_F = \hat{b} + \epsilon \hat{s}$).

Proof. Use the previous inequality (8) with velocity given by $b_F - \epsilon \nabla \log \hat{\rho}$. □

In fact, one can use this to tune the optimal value of ϵ to be

$$\epsilon^* = \sqrt{\frac{\text{excess risk in } \hat{b}}{\text{excess risk in } \hat{s}}}. \tag{18}$$

Note that if b is learned perfectly, this implies that the optimal $\epsilon = 0$, which corresponds to the ODE case and matches our instinct from the KL bound control in (8).

It can be shown via explicit constructions that these interpolants can replicate SB-DMs, denoising methods, and even rectified flows, often combining nice properties from each (for instance, smoothing out some singular behaviors of the velocity at the endpoints).

5 Possible extensions

- The statistical error analysis above assumes a perfect solution to the PDEs, which are solved via discretizations in practice for both the ODE and SDE approach. [2][3] conduct the analysis for the finite-time case. That said these suffer from dimension quadratically, while empirical results show linear dependence. Can this be shown?
- Work has been done to extend this framework to infinite dimensions in [4], but restricted to Hilbert spaces with the same topology on both ends. Is it possible to come up with a framework where distributions over different spaces (euclidean to graph for instance) can be connected?
- Singular behavior of the score and velocity fields at the endpoints seem to make the usual sample complexity analysis difficult. How can one deal with this to come up with sample complexity guarantees for the neural network learning these functions?

References

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